Shivam Dave

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Experience

Advanced Manufacturing Soft Robotics Researcher Summer 2026

ELIXIR Lab, Toronto Metropolitan University *Toronto, ON*

- Engineered robotic architectures to overcome actuation limitations by integrating **quantitative modeling, embedded hardware sensing, and closed-loop PID control**, improving system accuracy by **>25%** and reducing response latency
- Developed **3+ experimental prototypes** for dynamic robotic tasks using **real-time data acquisition and kinematic evaluation**, successfully validating adaptive grasping capabilities for advanced manufacturing applications
- Authored comprehensive technical documentation across **50+ trial iterations** to capture performance bottlenecks, presenting quantitative findings to cross-functional teams that successfully guided continuous hardware optimization

Emergency Department Clerical Associate 2024 to Present

William Osler Health Systems *Brampton, ON*

- Managed **real-time data acquisition and state tracking** across hospital systems during high-acuity triage scenarios, maintaining high-fidelity data integrity and achieving **100% compliance** for **120+ daily operations**
- Coordinated **concurrent processing requests** between **30+ clinical staff** per shift to address chaotic departmental workflows, optimizing resource allocation and significantly reducing information retrieval latency

Quantitative & Engineering Projects

Stock Market Analysis & Quantitative Research 2025 to 2026

Python, pandas, NumPy, SQL, Excel

- Analyzed **10+ years** of historical market data across **50+ equities** using **Python, pandas, and NumPy** to identify profitable market trends, successfully evaluating stock performance, volatility, and risk metrics like the Sharpe ratio
- Engineered automated ETL (Extract, Transform, Load) workflows in **SQL** to resolve unstructured data inefficiencies, cleaning **100,000+ data points** and accelerating data transformation by **30%** for advanced trend forecasting

Solar Panel Tracking PID Control System Simulation 2026

Python, MATLAB, Gazebo, Data Analysis, Control Systems

- Simulated **5 dynamic kinematic models** in **Gazebo and MATLAB** to address energy capture inefficiencies, optimizing tracking accuracy and achieving a projected **steady-state error reduction of <2%**
- Engineered a robust **closed-loop PID tracking architecture** to combat environmental disruptions, testing across **4 extreme conditions** (wind, saturation, dynamics) to minimize overshoot and optimize system settling time

Acoustic Phonoangiography Signal Processing System 2026

Python, NumPy, SciPy, Machine Learning, Streamlit

- Developed and deployed a **web-based Streamlit application** to provide accessible arterial stenosis screening, building an acoustic signal processing pipeline that successfully detects non-invasive vascular bruits
- Implemented **bandpass filtering, STFT, and Welch spectral estimation** using **SciPy** to eliminate ambient acoustic noise, successfully isolating **>200 Hz** turbulent flow frequencies and improving Signal-to-Noise Ratio (SNR)

Education

Toronto Metropolitan University Toronto, ON

Bachelor of Engineering, Mechatronics Engineering; Minor in Philosophy *2023 to Present*

- **Relevant Coursework:** Control Systems, Real-Time Digital Control Systems Design, Sensors and Instrumentation, Engineering Design, Embedded Systems, Manufacturing Processes, Numerical Analysis

Technical Skills & Certifications

Programming & Data Analysis: Python, NumPy, SciPy, SQL, MATLAB, C/C++, Linux, Git, JavaScript, React
Engineering Tools: Gazebo, SolidWorks, Fusion 360, AutoCAD, Inventor, Revit, Semantic Segmentation, CFD
Certifications: Six Sigma White Belt - *The Council for Six Sigma Certification*, IEEE Student Member, Financial Markets Certification - *Yale University Online*, Google AI Professional Certificate, Member of the Rehabilitation Engineering and Assistive Technology Society of North America, Medical Terminology Certificate - *Universal Class*